1 Advanced techniques for improving diversification of the structures

Recall the proposed workflow of the AIZyme project. Step 1 is served for generating structures without sequences by diffusion models. This step is required for generating a promising pool of secondary structures according to the reference structure (of `turboPETase` or similar enzyme). This pool of structures is further filtered by comparing the matching score in comparison to the structure of the reference molecule, and the best are kept and passed to predict sequences that correspond to these structures. This is done by employing a deep learning model `ProteinMNPP`, generating several possible sequences for each structure. Further, all these predicted sequences are passed to Step 2, involving the `AlphaFold`, for predicting their secondary structures. The bunch of produced structures is passed to Step 3, where these two set of structures, the initial one and the these generated by `AlphaFold`, are compared mutually. Here, filtering by removing bad candidates is applied, and candidate with best parameters are selected. For this purpose, various metrics are used such as RMSD score ranking. The obtained/survived sequences are sent to Laboratory testing (AI-zyme), and the procedure is repeated again from over.

The above designed workflow presents the backbone of our "de novo" enzyme generation, relying on several core components such as use of several powerful tools to converge towards a pool of promising (mimic) enzymes (`AlphaFold`, neural network `ProteinMPNN`, and diffusion models [1,2,3]) which will further be processed in the laboratory hopefully confirming their catalytic properties. As this piece of puzzle is most time-demanding but also most expensive task, one has to ensure—or at least increase the possibility—for selecting those structurally and semantically different secondary structures but still sufficiently realistic ones (in terms of catalytic characteristics) that would present new enzymes with a new functionality subsequently proven in the lab testing.

As already mentioned, the first idea incorporates several filtering mechanisms to diversify the process of secondary structure-guided novel protein sequence generation. These filtering mechanisms rely on employing measuring distances in the graph space (secondary structures can be seen as 3D spatial graphs) by applying Root Mean Square (RMS) distance, or Root Mean Square Deviation (RMSD), as a metric to quantify similarity between two structures by measuring the average difference between their structures or a referent structure. In that way, it will be given a rough estimate on how graphs really match each other in the search space of (secondary) structures. The procedure can be seen as an explicit way of filtering by taking absolute positions of structures in the search space relative to the referent enzyme (e.g., turboPETase). However, this usually may not be sufficient and is dependent of the threshold value that indicate if two structures are similar enough so that non-referent structure can be removed. Based on it, we will study the sensibility of `AlphaFold` w.r.t. input sequences and their score of similarity/alignment and similarity of the output structures (secondary structure of sequences). This is currently weakly explored in the literature and might be done by considering various measures of sequential similarity between multiple sequential structures. Among many, the well-established longest common subsequence (LCS) problem [4] seeks for object called sequences for detecting longest possible common motifs between pool of sequences, relying on structural similarities between input sequences. There are many practical variants of the LCS problem from the literature—the LCS arc-preserving problem, longest filled subsequence problem, constrained LCS problem, or gapped longest subsequence problem [5,6,7] to name a few. Their outcome is represented by more realistic patterns common to all input sequences. In that sense, we could empirically measure the (structural) similarity of sequences before passing them to `AlphaFold`. If the relative LCS score is high, meaning the sequences can be well-aligned, it will produce a red flag for increasing the diversification of sequences. Another measure that could serve for this purpose is the famous *Multiple Sequence Alignment* (MSA) problem [8,9] and its score reflecting on the size of common patterns embedding. If the need for sequential diversification is detected, the current flow will be moved back to the deep learning model (`ProteinMNPP`) sending a request for generating more diversified sequences.

Another step in the proposed workflow which could potentially deal with an increased diversification of secondary structures before going to the lab, is the subsequent step when the `AlphaFold` has generated candidate structures of the sequences, which are then filtered out by RMSD score or another one w.r.t. the reference structure. The idea is to analyze these generated structures deeply. In essence, several measure indicators will be incorporated for each of these structures: pocket volume, Catalytic residue geometry, and stability indicators—referring to different domination-based optimization invariant in graphs, and graph topology features such as centrality measure, radius of graphs, density, etc. The size of minimum dominating set [10,11], the maximum (quasi-) clique [12] are known invariant that carry on important aspects in biology, protein stabilization, etc. Note that the later two problems represent NP-hard combinatorial optimization problems, thus, for large-scale graphs computationally challenging to be examined. However, this does not give a substantial issue, as these problems are well studied in the literature where various effective meta-heuristic approaches are designed, able to reach close-to-optimal solutions within reasonable runtimes for large-scale graphs [13,14]. Here one has to deal with controlling these solvers with accordance to the allowed (time and memory) resources, therefore will be appropriately tuned by, for example, `Optuna` tool [16]. One idea is to perform the process of feature extraction on the HPC (VEGA) concurrently in parallel fashion, speeding up the procedure significantly. Having all these (reasonably cheap) features for each generated structure is expected to be important in identifying group of similar structures, revealing/detecting those bad ones that can be replaced by improving diversity of pool of structures. Let us assume for each structure s, a set of relevant (d) features $x_s \in \mathbb{R}^d$ is extracted. We tend to explore the idea of consensus problem based on employing different clustering techniques in Machine learning, i.e., unsupervised learning frequently used in bioinformatics—in detecting clusters of proteins with same functionality [15] in large PPI networks. Further, if by S we denote a set of all structures dealing with, one can use clustering techniques such as k-means or a hierarchical clustering to cluster structures in S into clusters of similar structures $(\mathcal{C}_1, \ldots \mathcal{C}_k)$ w.r.t. similar feature values; one would here ask for the existence of highest possible number k of clusters providing a good clustering (e.g., by reporting Silhouette coefficient). For each cluster $\mathcal{C}_i$, one could uniformly sample a portion (p) of structures which would be kept for promising candidates, whilst the other structures omitted and new ones regenerated by the diffusion model. These new ones would be passed through the `ProteinMPNN` and subsequently `AlphaFold`, until a predefined number of secondary structures is generated, likely more diversified than the starting one. Afterwards, it will be moving to Step 3 of the workflow.

Research question: Do designed enzyme mimics reproduce native catalytic geometry? TODO: a paragraph about it should be written...

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